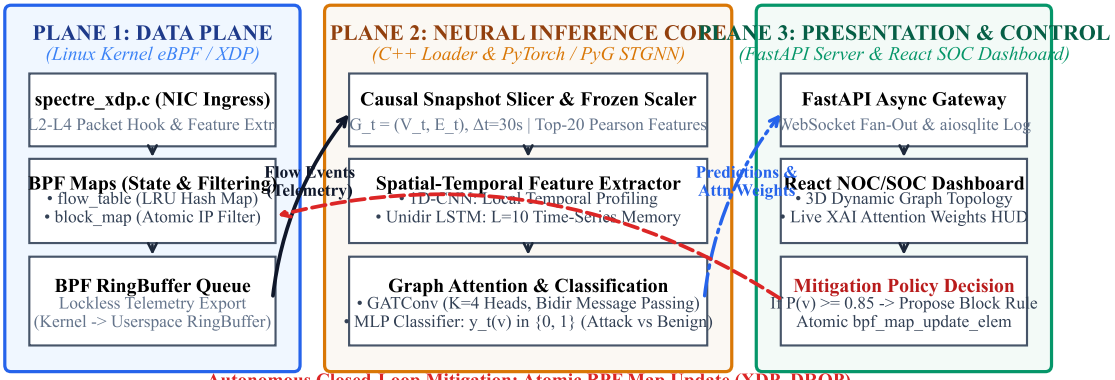

PLANE 1: DATA PLANE
(Linux Kernel eBPF / XDP)
spectre_xdp.c (NIC Ingress)
L2-L4 Packet Hook & Feature Extr
BPF Maps (State & Filtering)
• flow_table (LRU Hash Map)
• block_map (Atomic IP Filter)
BPF RingBuffer Queue
Lockless Telemetry Export
(Kernel -> Userspace RingBuffer)
PLANE 2: NEURAL INFERENCE CORE
(C++ Loader & PyTorch / PyG STGNN)
Causal Snapshot Slicer & Frozen Scaler
G_t = (V_t, E_t), Δt=30s | Top-20 Pearson Features
Spatial-Temporal Feature Extractor
• TD-CNN: Local Temporal Profiling
• Unidir LSTM: L=10 Time-Series Memory
Graph Attention & Classification
• GATConv (K=4 Heads, Bidir Message Passing)
• MLP Classifier: y_t(v) in {0, 1} (Attack vs Benign)
PLANE 3: PRESENTATION & CONTROL
(FastAPI Server & React SOC Dashboard)
FastAPI Async Gateway
WebSocket Fan-Out & aiosqlite Log
React NOC/SOC Dashboard
• 3D Dynamic Graph Topology
• Live XAI Attention Weights HUD
Mitigation Policy Decision
If P(v) >= 0.85 -> Propose Block Rule
Atomic bpf_map_update_elem
Flow Events (Telemetry)
Predictions & Attn. Weights
Autonomous Closed-Loop Mitigation: Atomic BPF Map Update (XDP_DROP)
Primary Data Plane Flow [Empirically Validated in Offline Benchmark]
Telemetry & Visualization Flow [Functional Integrated Prototype]
Actuation Control Pathway*